PRODUCT REQUIREMENTS DOCUMENT

Voice Concierge

for The Meridian Casino & Resort

Software Engineer Take-Home Assessment

CONFIDENTIAL

Overview

The Meridian Casino & Resort is a luxury destination in Las Vegas. We need a voice-based concierge system that allows guests to ask questions and receive instant, spoken answers about our property, amenities, and services. The system should capture questions we can't answer so our team can continuously improve coverage.

This assessment consists of a core assignment and a bonus challenge. The core assignment focuses on building the voice agent and backend infrastructure. The bonus challenge adds an admin panel for content management.

Problem Statement

Our concierge desk and phone lines are overwhelmed with repetitive questions — hours, restaurant reservations, game availability, room types. Guests expect instant answers at 3 AM when they want to know if the poker room is open. Meanwhile, we have no visibility into what guests are asking that we can't help with, meaning we miss opportunities to improve our service.

Goals

Core Goals

- Provide 24/7 instant voice support for common guest questions
- Capture unanswered questions for later review
- Maintain the luxury, personalized feel guests expect from The Meridian
- Provide a playground interface to test the voice concierge

Bonus Goals

- Allow staff to manage FAQ content through an admin panel
- Review and convert unanswered questions to FAQs
- Configure voice personality through the admin interface
- Integrate the playground into the admin panel

User Personas

Guest (Core)

- Staying at the hotel or visiting the casino
- Wants quick, accurate information without waiting on hold

- Expects a polished, professional experience befitting a luxury property
- May ask about gaming, dining, accommodations, entertainment, or local partners

Concierge Manager (Bonus)

- Manages FAQ content and ensures accuracy
- Reviews questions the system couldn't answer
- Adds new information based on guest inquiries
- Configures voice personality to match brand standards
- Tests the concierge experience before deploying changes

Functional Requirements: Core

Voice Concierge

ID	Requirement
VC-1	Guest can speak to the concierge through a web interface
VC-2	Concierge understands natural language questions
VC-3	Concierge searches knowledge base to find relevant answers
VC-4	Concierge responds verbally with the answer when found
VC-5	When no answer exists, concierge apologizes gracefully and confirms the question has been noted
VC-6	Tone should be warm, professional, and consistent with luxury hospitality

Knowledge Base

ID	Requirement
KB-1	Store FAQ items covering all property information
KB-2	Support natural language matching (not just keywords)
KB-3	Store unanswered questions with timestamp
KB-4	Track frequency of each unanswered question
KB-5	Seed with provided Meridian property information

Backend API

The backend should expose endpoints for the voice agent to interact with the knowledge base.

ID	Requirement
API-1	Endpoint to search FAQ items by question/query
API-2	Endpoint to record an unanswered question
API-3	FAQ search should return the best matching answer or indicate no match found

Playground Interface

A simple web interface to test the voice concierge. This can be the LiveKit Agents Playground or a custom implementation.

ID	Requirement
PG-1	Web-based interface accessible via browser
PG-2	Microphone input to speak to the concierge
PG-3	Audio output to hear concierge responses
PG-4	Visual indicator showing connection/conversation status
PG-5	Ability to start and end conversations

Note: The LiveKit Agents Playground (https://agents-playground.livekit.io) can be used as the playground interface, or you may build a custom one.

Functional Requirements: Bonus

The following requirements are optional bonus challenges. Complete the core requirements first.

Voice Configuration

ID	Requirement
VX-1	System supports 4 distinct voice options
VX-2	Admin can select active voice from admin panel
VX-3	Voice change takes effect immediately for new conversations
VX-4	Admin can preview each voice before selecting

Voice Options

Voice ID	Name	Description
1	James	Male, mature, warm British accent. Professional and refined.
2	Sofia	Female, friendly, subtle European accent. Welcoming and elegant.
3	Marcus	Male, American, confident and energetic. Modern and approachable.
4	Elena	Female, American, calm and reassuring. Sophisticated and clear.

Note: Actual voice implementation will depend on TTS provider capabilities. These descriptions represent the target personality for each voice.

Admin Panel

A React-based admin panel for managing the concierge system.

FAQ Management

ID	Requirement
AP-1	View all FAQ items
AP-2	Add new FAQ item
AP-3	Edit existing FAQ item
AP-4	Delete FAQ item

Unanswered Questions

ID	Requirement
AP-5	View unanswered questions queue
AP-6	See frequency count for each unanswered question
AP-7	Convert unanswered question to FAQ by providing an answer
AP-8	Dismiss irrelevant questions

Voice Configuration UI

ID	Requirement
AP-9	View available voice options with descriptions
AP-10	Select active voice for the concierge
AP-11	Preview button to hear sample from each voice
AP-12	Display which voice is currently active

Integrated Playground

ID	Requirement
AP-13	Embedded voice interface to test the concierge within the admin panel
AP-14	Playground uses current voice and FAQ configuration
AP-15	Allows admin to have full voice conversation with the concierge
AP-16	Clearly labeled as "Test Mode" or "Playground"

Property Information (Seed Data)

General Information

Topic	Details
Property Name	The Meridian Casino & Resort
Location	Las Vegas Strip, Nevada
Casino Hours	24 hours, 7 days a week
Hotel Check-in	4:00 PM (early check-in subject to availability)
Hotel Check-out	11:00 AM (late checkout available for suite guests)
Valet Parking	Complimentary for hotel guests, $25 for visitors
Self Parking	Free for all guests
Dress Code	Smart casual throughout property; formal attire required at Aurelia restaurant and Eclipse Lounge
Age Requirement	21+ for casino floor and bars; all ages welcome in hotel and family restaurants
Wi-Fi	Complimentary throughout property; premium high-speed available in suites

Casino Gaming

Game/Area	Details
Slot Machines	Over 2,000 machines ranging from $0.01 to $1,000 per spin. Includes latest video slots, classic reels, and progressive jackpots. Largest jackpot currently at $4.2 million.
Blackjack	40 tables, minimums from $25 to $10,000. Single deck, 6-deck, and Spanish 21 available.
Poker Room	24/7 operation. Texas Hold'em (No Limit and Limit), Omaha, and Seven Card Stud. Daily tournaments at 11 AM and 7 PM. $200 buy-in for daily tournaments.
Roulette	12 tables including American, European, and French roulette. Minimums from $15.
Baccarat	8 tables in main casino, private high-limit salon with 4 tables. Minimums from $50, high-limit from $5,000.
Craps	6 tables, minimums from $15. Lessons available daily at 10 AM.
Sports Book	80-seat theater with 40-foot screen. Full bar service. Mobile betting available throughout property via Meridian app.
High Limit Salon	Private gaming area with dedicated hosts. By invitation or $50,000 credit line. Private restrooms, complimentary dining, personal butler service.

Players Club	Meridian Rewards program. Earn points on all play. Tiers: Gold, Platinum, Diamond, and invite-only Black.

Accommodations

Room Type	Details
Deluxe Room	450 sq ft. King or two queens. City or pool view. From $299/night.
Premier Room	550 sq ft. King bed, sitting area. Strip view. From $449/night.
Luxury Suite	900 sq ft. Separate living room, marble bathroom with soaking tub. From $799/night.
Penthouse Suite	1,800 sq ft. Two bedrooms, dining room, butler's pantry, private terrace. From $2,500/night.
Chairman's Villa	4,500 sq ft. Three bedrooms, private pool, personal chef available, 24-hour butler. By inquiry only.
Accessible Rooms	Available in all categories. Roll-in showers, lowered amenities, visual alerts. Request at booking.

On-Property Restaurants

Restaurant	Details
Aurelia	Fine dining. Modern French cuisine by Chef Marcus Webb (2 Michelin stars). Tasting menu $285 per person, à la carte available. Dinner only, 6 PM - 10 PM. Reservations required. Formal attire.
Silk Road	Pan-Asian cuisine. Sushi bar, robata grill, dim sum. Open 11 AM - 11 PM daily. Reservations recommended for dinner. Smart casual. Prices $40-80 per person.
The Steakhouse	Classic American steakhouse. Dry-aged beef, fresh seafood. Open 5 PM - 11 PM. Reservations recommended. Smart casual. Prices $70-150 per person.
Café Meridian	Casual all-day dining. Breakfast buffet 7-11 AM ($45). Lunch and dinner menu until midnight. No reservations needed.
Pool Bar & Grill	Poolside casual. Burgers, salads, cocktails. 10 AM - 6 PM (seasonal). Hotel guests only.

Bars & Lounges

Venue	Details
Eclipse Lounge	Rooftop bar with Strip views. Craft cocktails, champagne menu. Open 5 PM - 2 AM. Formal attire after 8 PM.
The Vault	Whiskey and cigar lounge. 400+ whiskey selections. Humidor with premium cigars. Open 4 PM - 2 AM. 21+.
Casino Floor Bars	6 bars throughout casino. Complimentary drinks for active players. Full service available.

Entertainment & Amenities

Amenity	Details
Meridian Spa	Full-service spa and salon. Massage, facials, body treatments. Couples suites available. Open 8 AM - 8 PM. Book 24 hours in advance recommended.
Fitness Center	24-hour access for hotel guests. Personal training available. Peloton bikes, free weights, yoga studio.
Pool Complex	Three pools including adults-only infinity pool. Cabanas from $300/day (includes $100 food credit). Open 8 AM - 8 PM.
Meridian Theater	1,200-seat venue. Current residency: internationally acclaimed artists (check website for schedule). Tickets from $95. VIP packages available.
Nightclub - NOVA	Premier nightclub. Open Friday and Saturday, 10:30 PM - 4 AM. Cover $50 general, tables from $2,000. Guest list available for hotel guests.

Celebrations & Special Events

Service	Details
Weddings	Multiple venues from intimate chapel to grand ballroom. Packages from $5,000 to $150,000. Wedding coordinator included in all packages.
Private Events	Meeting rooms and ballrooms from 500 to 15,000 sq ft. Full catering and AV services. Corporate rates available.
Birthday/Anniversary	Celebration packages include room upgrade, champagne, custom cake, and dinner credit. From $500 additional. 72-hour advance notice required.
Bachelor/Bachelorette	VIP packages including suite, nightclub table, pool cabana, and spa credits. Starting at $3,000. Must book 1 week in advance.

Partner Discounts (Show Room Key)

Partner	Discount
Carbone (Italian fine dining, 5 min away)	15% off food, priority reservations
Omega Mart (Immersive art experience)	$10 off admission
Helicopter Tours - Vegas Nights Aviation	20% off Strip tours, complimentary champagne
Top Golf Las Vegas	Free hour with 2-hour booking
Spa Aquae at JW Marriott	10% off treatments (when Meridian Spa fully booked)
Fremont Street Experience	VIP SlotZilla pass, skip the line

Example Conversations

Scenario 1: Gaming Question

Guest: "Is the poker room open right now?"

Concierge: "Yes, our poker room is open 24 hours a day, 7 days a week. We offer Texas Hold'em, Omaha, and Seven Card Stud. There's also a tournament starting at 7 PM tonight with a $200 buy-in if you're interested."

Scenario 2: Dining Recommendation

Guest: "What's your best restaurant?"

Concierge: "Our signature restaurant is Aurelia, featuring modern French cuisine by Chef Marcus Webb — it holds two Michelin stars. It's open for dinner from 6 to 10 PM and does require reservations and formal attire. If you'd prefer something more casual, Silk Road offers excellent Pan-Asian cuisine and The Steakhouse is perfect for classic American dining."

Scenario 3: Unknown Question

Guest: "Can I bring my dog to the hotel?"

Concierge: "I don't have information about pet policies at the moment, but I've noted your question for our team. You can also reach our front desk directly at extension 0. Is there anything else I can help you with?"

Scenario 4: Partner Discount

Guest: "Are there any good restaurants nearby you'd recommend?"

Concierge: "Absolutely. Carbone is an excellent Italian fine dining option about five minutes away, and Meridian guests receive 15% off and priority reservations — just show your room key. We also have three restaurants on property if you'd prefer to stay in."

Scenario 5: Celebration Planning

Guest: "I want to propose to my girlfriend this weekend. Can you help?"

Concierge: "How exciting! We have several special options. Our celebration packages start at $500 and include a room upgrade, champagne, and dinner credit. For something more elaborate, Eclipse Lounge offers private terrace reservations with stunning Strip views. I'd recommend speaking with our concierge desk to plan the perfect moment — would you like their direct number?"

Non-Functional Requirements

ID	Requirement
NF-1	Voice response should feel conversational and prompt
NF-2	System should run locally via Docker for evaluation
NF-3	Admin panel should be usable on desktop browsers
NF-4	Concierge tone should reflect luxury hospitality brand
NF-5	Voice selection change should apply without system restart

Out of Scope

- Actual reservation booking (just provide information)
- Payment processing
- Integration with property management systems
- User authentication for admin panel
- Guest-facing standalone website (guests use playground in admin for this exercise)
- Mobile app
- Multi-language support

Technical Constraints

Required for compatibility with existing infrastructure:

Core

Component	Allowed Technologies
Backend API	.NET, Java, Node.js/JavaScript, or Python (any framework)
Voice Agent	Python or JavaScript/TypeScript
Voice Infrastructure	LiveKit
Playground	LiveKit Agents Playground or custom implementation

Bonus

Component	Allowed Technologies
Admin Panel	React

All other technical decisions (database, search approach, LLM provider, architecture patterns, communication protocols) are at implementer's discretion.

Provided Resources

- LiveKit API credentials
- LLM API key
- STT/TTS service credentials (supporting multiple voices)

Deliverables

Core (Required)

- Working voice agent that answers FAQ questions
- Backend API with FAQ search and unanswered question recording
- Database seeded with Meridian property information
- Playground interface to test the voice concierge
- System launchable with a single command (e.g., docker-compose up)
- README with setup instructions
- Brief document explaining key technical decisions

Bonus (Optional)

- React admin panel with FAQ management
- Unanswered questions queue with convert-to-FAQ functionality
- Voice configuration UI with 4 voice options and preview
- Integrated playground within the admin panel

Success Criteria

Core (Required)

Area	Criteria
Voice Conversation	Can have natural voice conversation about Meridian property
FAQ Lookup	Returns accurate, relevant answers from knowledge base
Unknown Questions	Records unanswered questions to database
Playground	Can test the voice concierge through a web interface
Overall Experience	Feels like speaking with a knowledgeable, professional concierge

Bonus (Optional)

Area	Criteria
FAQ Management	Admin can view, add, edit, delete FAQ items
Question Queue	Admin can review and convert unanswered questions to FAQs
Voice Selection	Admin can choose between 4 voices, change applies immediately
Voice Preview	Admin can hear sample of each voice before selecting
Integrated Playground	Admin can test full voice experience within the admin panel

Questions? Contact us:

jenya@rocketdreams.co
+972 54-524-1285